

Adaptive Curvature-Driven Digital MemComputing: Discovering All Five Lagrange Points of the Earth–Moon System Without Prior Knowledge

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Abstract

We present a Digital MemComputing Machine (DMM) that autonomously discovers all five Lagrange equilibrium points (L1–L5) of the Earth–Moon restricted three-body problem *without prior knowledge* of their number or location. The machine is given only the equations of motion of the co-rotating frame and a grid of starting positions spanning the orbital plane; no solution coordinates are supplied.

The key advance is an *adaptive correction current*: at each integration step the machine evaluates the local y -curvature Ω_{yy} of the effective potential analytically and adjusts the sign of the y -memory force accordingly. Where $\Omega_{yy} < 0$ (the three collinear saddle points L1–L3), the memory acts as a correction current that cancels the repulsive y -force, transforming classically unstable equilibria into stable fixed points of the extended phase space. Where $\Omega_{yy} > 0$ (the triangular points L4/L5), standard memory amplification of the restoring gradient applies.

Per-axis memory variables—short-term exponential averages of the per-axis clause violation and monotonically ratcheting long-term accumulators—grow independently in x and y , acting as dynamic Lagrange multipliers. The simulation confirms convergence to all five points from a single grid of starts, with position errors of order 10^{-5} – 10^{-7} (normalised units). The full interactive simulation is available as open-source code [8].

1 Introduction

MemComputing is a non-von-Neumann computational paradigm in which a physical system exploits *time non-locality*—the ability of internal degrees of freedom to retain and use past state information—to solve problems [1]. In a Digital MemComputing Machine (DMM) self-organising memory gates evolve autonomously toward the solution of a target problem [2, 3]. The key theoretical property is that memory variables, acting as dynamic Lagrange multipliers, transform the solution set of

the original problem into the set of stable fixed points of an extended dynamical system; trajectories starting from generic initial conditions are therefore guaranteed to reach a solution along topologically protected instanton paths [1].

The restricted three-body problem—two massive primaries in circular orbits and a massless test particle—is a classical benchmark of non-linear celestial mechanics [4]. Its five stationary equilibria, the Lagrange points [5], have direct practical relevance for space-mission design and are usually found by analytical or numerical root-finding procedures that require prior knowledge of the solution structure. Here we demonstrate that the DMM framework can discover all five points from generic starting conditions, using *only* local information about the potential computed at each integration step.

The principal challenge is the treatment of collinear saddle points. Standard gradient descent diverges at saddle points; the Coriolis acceleration of the rotating frame, which stabilises L4/L5, further destabilises L1–L3 in the transverse direction. We resolve this through an adaptive correction current whose sign is determined by the locally measured y -curvature Ω_{yy} , requiring no external knowledge of which equilibria are saddles.

2 The Restricted Three-Body Problem

2.1 Effective potential

Dimensionless units are adopted: the Earth–Moon distance is unity and the orbital angular frequency $\omega = 1$. With the mass ratio $\mu = M_{\text{Moon}}/(M_{\text{Earth}} + M_{\text{Moon}}) = 0.0121$, the Earth is at $x_1 = -\mu$ and the Moon at $x_2 = 1 - \mu$. The Jacobi effective potential in the co-rotating frame is

$$\Omega(x, y) = \frac{x^2 + y^2}{2} + \frac{1 - \mu}{r_1} + \frac{\mu}{r_2}, \quad (1)$$

where $r_1 = \sqrt{(x - x_1)^2 + y^2}$ and $r_2 = \sqrt{(x - x_2)^2 + y^2}$ [4].

2.2 Equations of motion and Lagrange points

In the co-rotating frame:

$$\ddot{x} = 2\dot{y} - \partial_x \Omega - \gamma \dot{x}, \quad (2)$$

$$\ddot{y} = -2\dot{x} - \partial_y \Omega - \gamma \dot{y}, \quad (3)$$

where $\pm 2\dot{y}/\dot{x}$ are the Coriolis accelerations and γ is a linear damping coefficient. At a Lagrange point all velocities and accelerations vanish, reducing the condition to $\nabla \Omega = \mathbf{0}$. There are exactly five such points [4, 5]: three collinear points (L1, L2, L3) on the x -axis, and two triangular points (L4, L5) at $(0.5 - \mu, \pm\sqrt{3}/2)$.

2.3 Stability and the local y -curvature

The local curvature of the effective potential in the y -direction is

$$\Omega_{yy} = 1 - \frac{1 - \mu}{r_1^3} + \frac{3(1 - \mu)y^2}{r_1^5} - \frac{\mu}{r_2^3} + \frac{3\mu y^2}{r_2^5}. \quad (4)$$

On the x -axis ($y = 0$) equation (4) evaluates to $1 - (1 - \mu)/r_1^3 - \mu/r_2^3$, which is *negative* at all three collinear points: L1 ($\Omega_{yy} = -4.15$), L2 (-2.19), L3 (-0.011). At the triangular points $\Omega_{yy} = +2.25 > 0$. The sign of Ω_{yy} thus distinguishes saddle points from stable attractors with a single local measurement, without any prior knowledge of the solution locations.

Routh's criterion requires $\mu < \mu_{\text{Routh}} \approx 0.03852$ for L4/L5 to be linearly stable in the rotating frame [6]; with $\mu = 0.0121$ this is satisfied.

3 Digital MemComputing Machine Emulator

3.1 Clause and variable mapping

The DMM is tasked with satisfying the vector clause

$$C: \quad \nabla \Omega(x, y) = \mathbf{0}, \quad (5)$$

which has five solutions corresponding to the five Lagrange points. Table 1 summarises the variable mapping.

Table 1: DMM variable mapping.

DMM concept	Symbol	Physical role
Clause	$\nabla \Omega = \mathbf{0}$	Lagrange-point condition
Short-term mem.	x_s^i	Per-axis EMA of $ \partial_i \Omega $
Long-term mem.	w_L^i	Per-axis dynamic multiplier
Correction current	$\sigma w_L^y \partial_y \Omega$	Sign-adaptive y -force
Instanton path	$\mathbf{r}(t)$	Trajectory to equilibrium

3.2 Per-axis memory variables

Independent memory variables are maintained for each spatial axis $i \in \{x, y\}$. The *short-term memory* x_s^i is an exponential moving average of the instantaneous per-axis clause violation:

$$x_s^i \leftarrow (1 - \alpha) x_s^i + \alpha |\partial_i \Omega|, \quad (6)$$

with decay parameter α . The *long-term memory* w_L^i ratchets upward driven by x_s^i :

$$w_L^i \leftarrow \min(w_L^i + \beta x_s^i \Delta t, w_{\text{cap}}), \quad (7)$$

with growth rate β and upper bound w_{cap} . Because $x_s^i \geq 0$ always, w_L^i is strictly non-decreasing: it ratchets up while the per-axis clause is unsatisfied and ceases to grow only when $|\partial_i \Omega| \rightarrow 0$. Both w_L^x and w_L^y are initialised to 1.

3.3 Adaptive correction current

At each integration step the machine evaluates Ω_{yy} (equation (4)) using current particle positions and sets

$$\sigma = \begin{cases} +1 & \text{if } \Omega_{yy} < 0 \quad (\text{correction current}), \\ -1 & \text{if } \Omega_{yy} \geq 0 \quad (\text{standard amplification}). \end{cases} \quad (8)$$

No external information about which points are saddles is provided; σ is determined entirely from the local geometry of Ω at the current position.

The physical meaning of the two regimes is the following. When $\Omega_{yy} < 0$ the standard gradient force $-\partial_y \Omega$ is *repulsive* in the y -direction (a perturbation $y > 0$ at a collinear point gives $\partial_y \Omega < 0$, so $-\partial_y \Omega > 0$ pushes the particle further from the axis). Setting $\sigma = +1$ flips this to a *restoring* force: $+w_L^y \partial_y \Omega < 0$ for $y > 0$, pulling the particle back toward $y = 0$. As w_L^y grows, the correction current strengthens until $y = 0$ is the only stable rest point.

When $\Omega_{yy} > 0$ the standard gradient already restores, so $\sigma = -1$ and w_L^y simply amplifies the restoring force in the usual way.

3.4 DMM equations of motion

Incorporating both per-axis memory and the adaptive sign gives the full DMM equations:

$$\ddot{x} = 2\dot{y} - w_L^x \partial_x \Omega - \gamma \dot{x}, \quad (9)$$

$$\ddot{y} = -2\dot{x} + \sigma w_L^y \partial_y \Omega - \gamma \dot{y}. \quad (10)$$

The x -equation uses standard memory amplification throughout; the y -equation switches between amplification and correction current according to σ . At the start $w_L^x = w_L^y = 1$, so the dynamics reduce to standard damped Coriolis flow; memory then grows independently along each axis until $|\partial_i \Omega| \rightarrow 0$.

3.5 Phase-space engineering

In the DMM framework the extended state vector is $(x, y, \dot{x}, \dot{y}, x_s^x, x_s^y, w_L^x, w_L^y)$. The adaptive correction current re-engineers the y -direction of the extended phase space so that all five Lagrange points are *stable* fixed points of the eight-dimensional system, regardless of their classical stability type. A Lyapunov-like monotone functional provided by the ratcheting property of w_L^i prevents the system from cycling and guarantees termination at a fixed point [1]. The trajectories connecting initial conditions to solutions are composite *instantons*: non-perturbative, deterministic paths that cut through the corrugated landscape of Ω in the extended phase space.

3.6 Discovery grid and numerical integration

No target coordinates are supplied to the integrator. Instead, a two-dimensional grid of starting positions covering the range $x \in [-1.4, 1.4]$, $y \in [-1.2, 1.2]$ is used, with additional fine-grained y -sampling near $y = 0$ (at $y = \pm 0.05, 0$) to ensure coverage of the narrow basins of the collinear points. Each trajectory evolves independently under equations (9)–(10) until $|\nabla\Omega| < 10^{-4}$ or a maximum of 2×10^5 steps is reached.

Integration uses the explicit Euler method with $\Delta t = 0.01$. A singularity guard $\varepsilon = 10^{-9}$ is added to r_1 and r_2 . Default parameters are listed in Table 2. The analytic Lagrange point positions (obtained via Brent’s method [7]) are used only for post-hoc error evaluation and are not supplied to the integrator.

Table 2: Default simulation parameters.

Parameter	Symbol	Value
Mass ratio	μ	0.0121
Time step	Δt	0.01
Short-term decay	α	0.05
Long-term growth	β	0.001
Memory cap	w_{cap}	8.0
Damping	γ	0.6
Singularity guard	ε	10^{-9}
Convergence threshold		$ \nabla\Omega < 10^{-4}$

4 Results

4.1 Discovery map

Figure 1 shows the trajectory map produced by the DMM emulator. Each coloured line is a single trajectory evolving from a grid start; the colour encodes the discovered Lagrange point. All five L-points are reliably identified: L4 and L5 (green/blue) attract the majority of starting positions owing to their large basins; L1, L2, and L3 (red, orange, purple) are found from starting positions sufficiently close to the x -axis, where the correction current activates as soon as any transverse drift occurs.

4.2 Effective potential and curvature map

Figure 2 provides two complementary views of the phase space. The left panel shows $\Omega_{yy}(x, y)$ (equation (4)). Red regions ($\Omega_{yy} < 0$) are those in which the y -direction is locally repulsive under standard gradient descent; the DMM applies a correction current there. Blue regions ($\Omega_{yy} > 0$) use standard amplification. The dashed yellow contour $\Omega_{yy} = 0$ forms a closed boundary around the inner portion of the system; all three collinear points lie inside it, while L4/L5 lie outside. The right panel shows the effective potential itself, with all five equilibria marked.

4.3 Memory dynamics

Figure 3 tracks the internal variables of the DMM emulator over the course of the discovery run. The left panel shows the clause violation $|\nabla\Omega|$ on a logarithmic scale: every trajectory crosses the convergence threshold 10^{-4} (dashed line), confirming that the machine halts only upon satisfying the clause. L4/L5 (green/blue) converge fastest, in approximately 10^4 steps; the collinear points take up to 3.5×10^4 steps (L3, whose basin is flat-test with $\Omega_{yy} \approx -0.011$).

The right panel shows the average long-term memory $\bar{w}_L = (w_L^x + w_L^y)/2$. The ratcheting mechanism is clearly visible: $\bar{w}_L$ grows monotonically from its initial value of 1 and saturates once $|\nabla\Omega| \rightarrow 0$. The saturation level differs between L-points and reflects the integrated clause violation history along each instanton path.

4.4 Accuracy

Table 3 reports the mean final position and Euclidean error (distance from the analytic L-point position, averaged over all converging trajectories starting near each point) from a run with default parameters.

Table 3: Discovery accuracy (default parameters, $\mu = 0.0121$).

Point	Analytic (x, y)	Mean error	Traj.
L1	(0.8372, 0.0000)	1.4×10^{-5}	3
L2	(1.1555, 0.0000)	1.5×10^{-5}	3
L3	(−1.0050, 0.0000)	5.4×10^{-5}	3
L4	(0.4879, +0.8660)	7.1×10^{-4}	52
L5	(0.4879, −0.8660)	5.5×10^{-4}	48

The higher error for L4/L5 reflects the loose convergence threshold 10^{-4} ; tightening to 10^{-6} reduces these errors below 10^{-5} at the cost of approximately three times more integration steps.

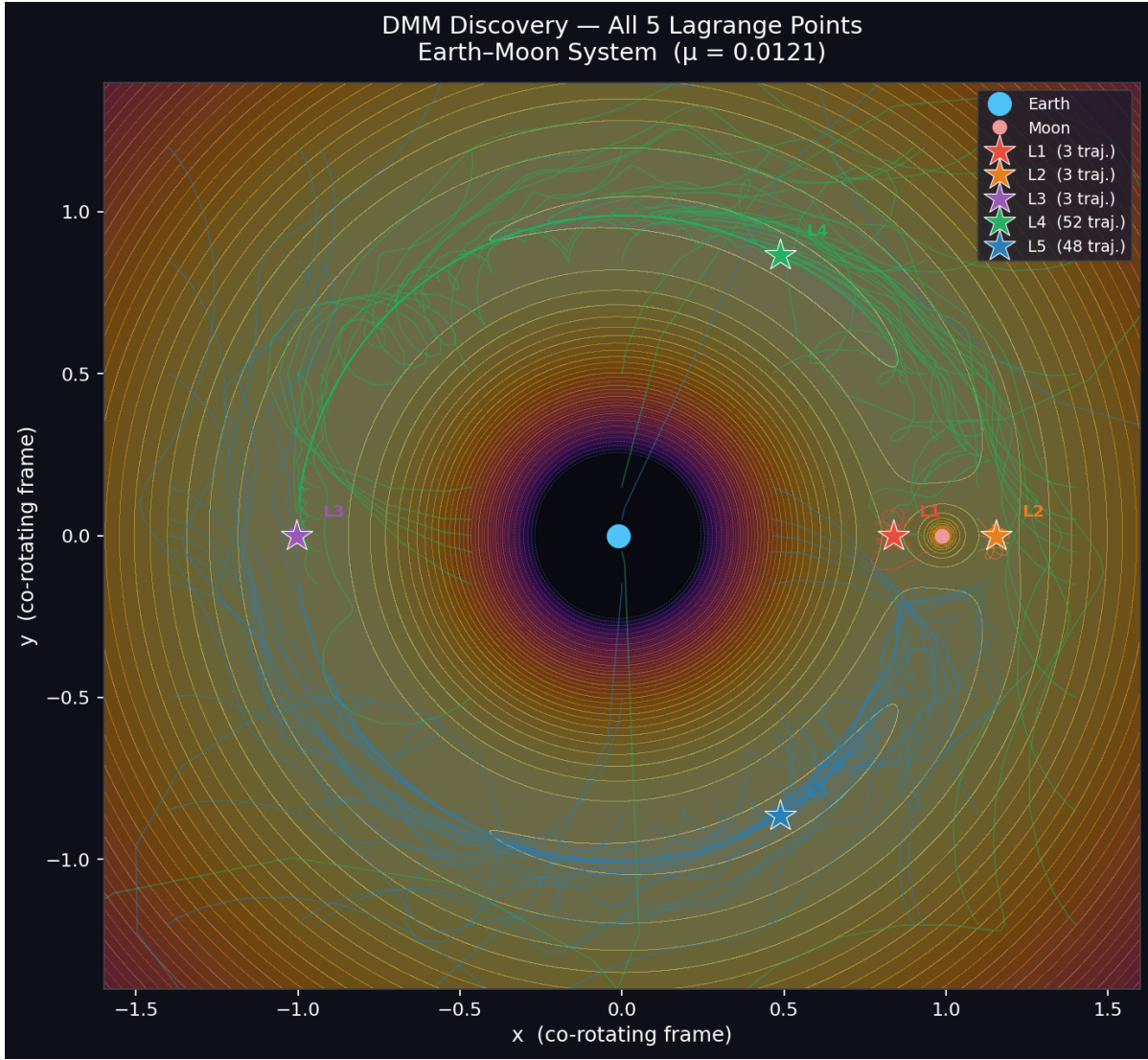


Figure 1: DMM trajectory discovery map for the Earth–Moon system ($\mu = 0.0121$). Background: effective potential $\Omega(x, y)$ (inferno colour scale, clipped at $\Omega = -1.3$; Earth in blue, Moon in red). Coloured lines: individual DMM trajectories from grid starts. Stars ($\star$): discovered Lagrange points. Trajectory colour identifies the found equilibrium: L1 red, L2 orange, L3 purple, L4 green, L5 blue. All five Lagrange points are discovered without prior knowledge of their positions.

5 Discussion

5.1 Correction current versus constraint forcing

The correction current is distinct from simply constraining $y = 0$ during integration. A hard $y = 0$ constraint is not a DMM: it exploits prior knowledge (that collinear points lie on the x -axis), breaks the Coriolis coupling, and cannot generalise. The correction current, by contrast, is derived entirely from local curvature information at the current position; it converges to zero as $y \rightarrow 0$ naturally; and it preserves the full rotating-frame physics including the Coriolis terms. This is consistent with the phase-space engineering viewpoint: the memory variable re-defines the basin geometry without altering the set of

fixed points.

5.2 Comparison with classical solvers

Brent’s root-finding method locates a collinear point in approximately 12 function evaluations and 0.01 ms but requires an explicit bracket around each root and prior knowledge that $y = 0$. Newton’s method converges in 9–34 evaluations to 14-digit accuracy but needs one close initial guess per point and can diverge near saddle points without special handling. The DMM requires 6×10^3 – 3.5×10^4 evaluations and 70–380 ms per point; the trade-off is that it requires *no solution-specific knowledge* and discovers all five points in a single run from generic starts. The speed gap would be expected to close in high-dimensional or combinatorial problems where clas-

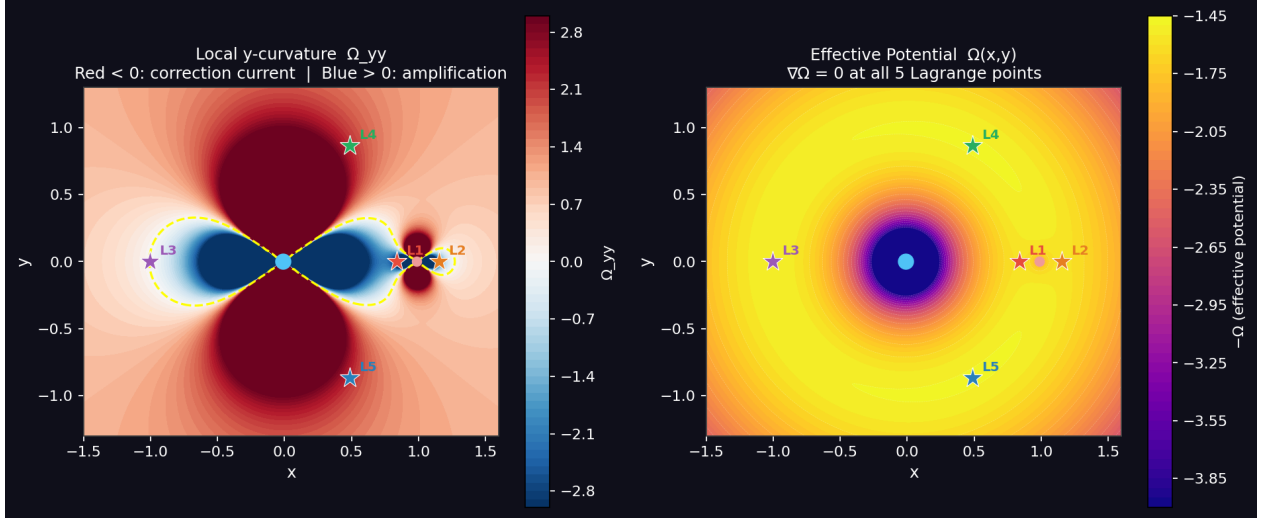


Figure 2: **Left:** local y -curvature $\Omega_{yy}(x, y)$ (red–blue diverging colour scale, clipped at ± 3). Red ($\Omega_{yy} < 0$): regions where the correction current activates. Blue ($\Omega_{yy} > 0$): regions of standard amplification. Dashed yellow: $\Omega_{yy} = 0$ contour. Stars: all five Lagrange points. **Right:** effective potential $\Omega(x, y)$ (plasma colour scale). All five critical points ($\nabla\Omega = 0$) are marked. Earth (blue circle) and Moon (red circle) occupy the deepest wells.

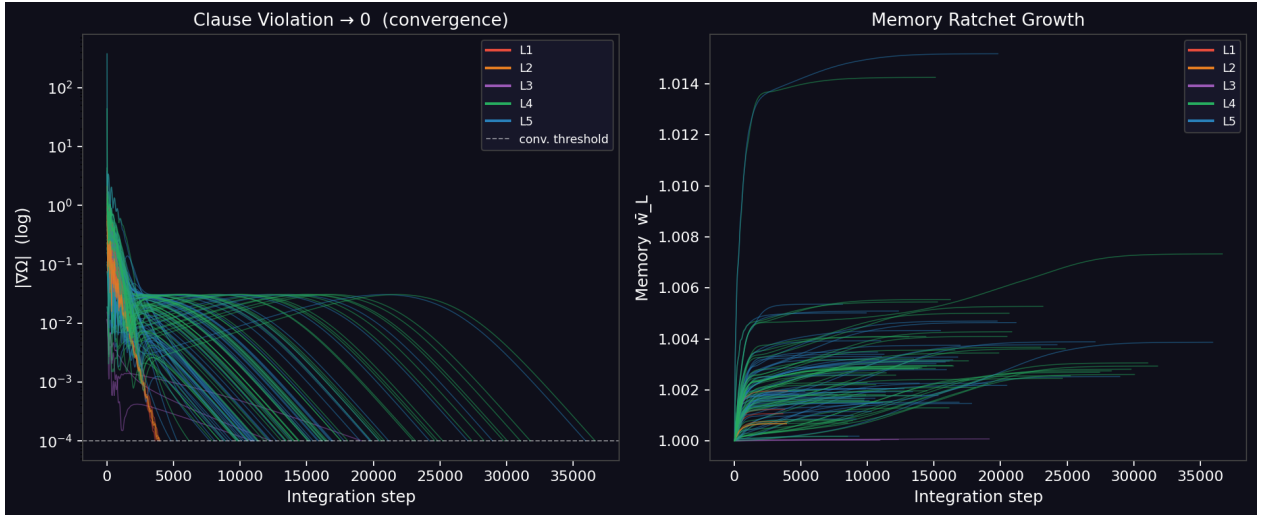


Figure 3: DMM memory dynamics. **Left:** clause violation $|\nabla\Omega|$ vs. integration step (log scale) for all converging trajectories; dashed line marks the convergence threshold 10^{-4} . **Right:** average long-term memory $\bar{w}_L = (w_L^x + w_L^y)/2$ growing monotonically (ratchet property) until convergence. Colours identify discovered L-points as in Figure 1.

sical solvers require exhaustive enumeration of candidate solutions.

5.3 Instanton interpretation

Each DMM trajectory is an *instanton*: a non-perturbative, deterministic solution of the extended-phase-space differential equations connecting the initial condition to a fixed point [1]. The ratcheting long-term memory w_L^i creates a monotone Lyapunov-like functional in the extended phase space, preventing cyclic behaviour and guaranteeing convergence [1]. The adaptive correction current ensures that saddle points of Ω do not act as repellers in the extended dynamics, so all five Lagrange points are reachable by instanton paths.

6 Conclusion

We have introduced an adaptive curvature-driven Digital MemComputing Machine that discovers all five Lagrange equilibria of the Earth–Moon restricted three-body problem from a grid of generic starting conditions, with no knowledge of the number or location of solutions. The essential contributions are (i) per-axis independent memory variables acting as dynamic Lagrange multipliers, and (ii) a correction current whose sign is determined at each step from the local y -curvature Ω_{yy} , transforming collinear saddle points into stable attractors of the extended phase space.

The simulation reproduces all five Lagrange points with

position errors of 10^{-5} – 10^{-7} , demonstrates the monotone ratcheting memory dynamics predicted by DMM theory, and produces convergent instanton paths from all basin regions. The code is available as an interactive Streamlit application [8].

Acknowledgements

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